Migration of applications from ATMEGA328 family to STM32C0 series microcontrollers

Introduction

This application note provides some guidelines and methodology to migrate easily an Arduino® application from the ATMEGA328 family to the STM32C0 series platform. It groups together all the most important information and lists the main aspects that must be addressed, providing guidelines on the hardware, firmware and peripheral migration.

For a better understanding, the user needs to be familiar with STM32 microcontrollers.

For additional information, refer to the documents in Table 1.

1 General information

This document applies to all STM32C0 series devices. All these products are Arm®-based microcontrollers.

Note: Arm® is a registered trademark of Arm® Limited (or its subsidiaries) in the US and/or elsewhere.

Table 1. Document and website references

Reference	Document
RM0490	STM32C0x1 advanced Arm®-based 32-bit MCUs(1)
DS13866	Arm® Cortex®-M0+ 32-bit MCU, 32 KB flash, 6 KB RAM, 2 x USART, timers, ADC, communication interface I/Fs, 2-3.6V(1)
DS13867	Arm® Cortex®-M0+ 32-bit MCU, 32 KB flash, 12 KB RAM, 2x USART, timers, ADC, communication interface I/Fs, 2-3.6V (1)
PM0223	Cortex®-M0+ programming manual for STM32C0, STM32L0, STM32G0, STM32WL and STM32WB series(1)
AN5373	Getting started with STM32C0 series hardware development (1)
AN4894	EEPROM emulation techniques and software for STM32 microcontrollers
STM32CubeProg	https://www.st.com/stm32cubeprog
STM32duino	Wiki: https://github.com/stm32duino/wiki/wiki Forum: https://www.stm32duino.com/ Arduino_Core_STM32:-https://github.com/stm32duino/Arduino_Core_STM32(2)
Getting started	https://github.com/stm32duino/wiki/wiki/Getting-Started#installing-stm32-cores (2)
Open Bootloader library	

1. *Available at www.st.com. Contact STMicroelectronics when more information is needed*
2. *This URL belongs to a third party. It is active at document publication, however STMicroelectronics shall not be liable for any change, move or inactivation of the URL or the referenced material.*

2 STM32C0 series overview

The STM32C0 series include all ATMEGA328 standard peripherals like SPI, UART (see Table 5 for more details). But also a set of peripherals with advanced features and optimized power consumption level like:

The STM32C0 devices include a set of peripherals with advanced features and optimized power consumption compared to the ATMEGA328:

- 32-bit CPU with CPU frequency at 48 MHz
- DMA
- 12-bit ADC

3 Hardware migration

3.1 Pinout compatibility

STM32C0 devices use a different system of power distribution (single supply pair), with the merging of V_{DDA} and V_{DD}. The STM32C0 has better GPIO density than the ATMEG328 (5 V supply voltage).

Due to the significant difference between ATMEG328 and STM32C0, there is no pin to pin compatible. In case of replacement, the PCB routing needs to be reworked or use an STM32C031C6 discovery kit (see Section 3.2).

Table 2. **Additional IOs for STM32C0 vs ATMEG328**

Package pin count	GPIO number in STM32C0	GPIO number in ATMEG328	Difference I/Os
28	26	23	+3
32	30	25	+5

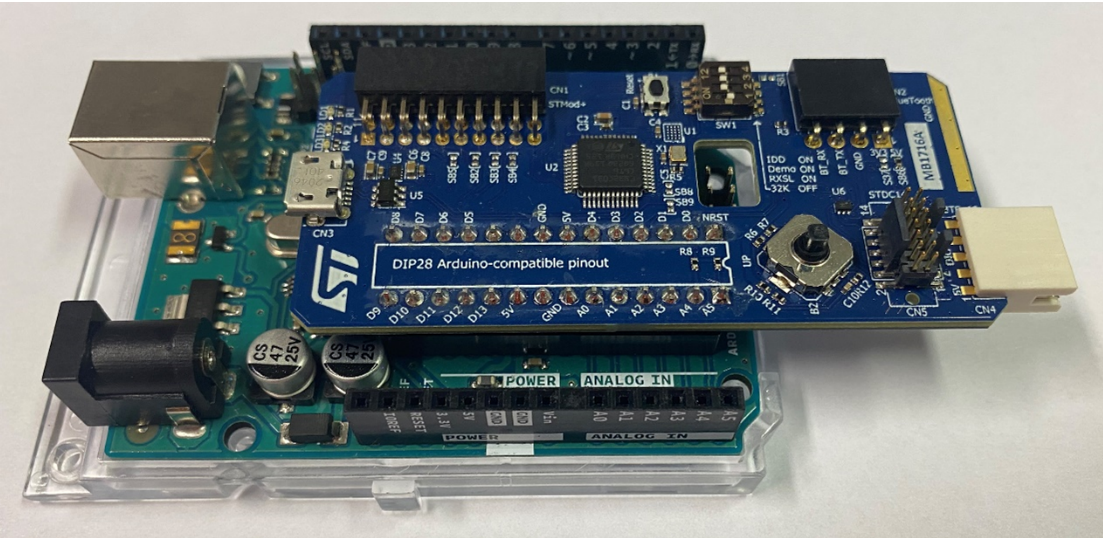
Table 3. **Package type**

Package pins	STM32C0 family	ATMEG328
8	SO8N	-
12	WLCSP	-
20	TSSOP/UFQFPN	-
28	UFQFPN	VQFPN / SDIP
32	UFQFPN / LQFP	VQFPN / TQFN
48	UFQFPN / LQFP	-

3.2 Board migration

To help the user to migrate easily from the ATMEGA328 to the STM32C0, a special STM32C0316-DK discovery board has been designed with a DIP28 Arduino®-compatible pinout. The Arduino® Uno board or homemade PCB can be easily plug-in at the place of the ATMEGA328 as shown in Figure 1:

Figure 1. **Arduino® uno board with STMC0316-DK**



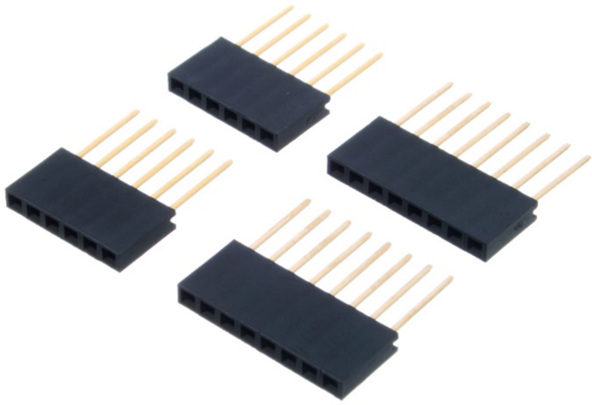
The two MCUs do not use the same pin name. Table 4 shows a comparison between the two architectures.
The biggest difference is concerning the ADC reference voltage pin. In the STM32 discovery board VREF is connected to VDD, so it can be impossible to use it in the Arduino® board or homemade PCB. Otherwise, with the help of STM32duino, the Arduino® core supports the STM32C0.

Table 4. Arduino® pin comparison

Arduino® pin	STM32C0	ATMEGA328
RESET	PF2-NRESET	PC6
A5	PB10	PC5
A3	PA9	PC3
A2	PA10	PC2
A1	PA5	PC1
A4	PB11	PC4
A0	PA4	PC0
AREF	Not connected	AREF
D19/SCL	PB11	PC5
D18/SDA	PB10	PC4
D13	PA1	PB5
D12	PA6	PB4
D11	PA7	PB3
D10	PB0	PB2
D9	PA8	PB1
D8	PD3	PB0
D7	PD2	PD7
D6	PD1	PD6
D5	PD0	PD5
D4	PA0	PD4
D3	PC7	PD3
D2	PC6	PD2
D1	PA2	PD1
D0	PA3	PD0

If the user use shield like Arduino® or ST extension board (compatible with Arduino® R3 connector), it is necessary to raise the header with a shield-stacking header.

Figure 2. Arduino® shield-stacking header



Note: If the application needs to use A4 and A5 as ADC input, it is mandatory to unsolder R8 and R9 resistors. These pull-up resistors are used as I^2C as default.

4 Boot mode selection

The boot configuration of the STM32C0 is based on the STM32 M0+ core products.

In the ATMEGA328 family the software can boot only from flash memory or bootloader while STM32C0 series permits to locate the BOOT vector in flash memory, system memory (boot loader) or RAM, based on the Table 4.

In fact it relocates the boot memory start address, for example if the user chose to boot from the main flash memory, this memory area is aliased in the boot memory space (0x0000 0000), but still accessible from its original memory space (0x0800 0000). It is reciprocal to other boot area.

The ATMEGA328 does not integer a system bootloader like in the STM32C0 (bootloader programmed during production) but something close to the open bootloader in other words the user need to add is own code. If the user wants to know more about the open bootloarder on the STM32: Open Bootloader library

A feature to check if the device is virgin has been implemented on the STM32C0 series. If the BOOT0 pin is defining main flash memory as the target boot area and after loading option byte, the flash memory interface checks if the first location of the main memory is programmed and return the result on FLASH_ACR register.

Table 5. Boot mode configuration

Boot mode configuration					Selected boot area
BOOT_LOCK bit	nBOOT1 bit	BOOT0 pin	nBOOT_SEL bit	nBOOT0 bit	
0	X	0	0	X	Main flash memory
0	1	1	0	X	System memory
0	0	1	0	X	Embedded SRAM
0	X	X	1	1	Main flash memory
0	1	X	1	0	System memory
0	0	X	1	0	Embedded SRAM
1	X	X	X	X	Main flash memory forced

5 Peripheral migration

5.1 STM32 product cross-compatibility

The STM32C0 and the ATMEGA328 shared the same set of peripherals with a few exceptions. The STM32C0 has: four timer 16 bit, DMA, RTC, I^{2}C. But it does not have a comparator like ATMEGA328.

Table 6. Peripheral summary of STM32C0 series and ATMEGA328

Peripheral		STM32C0 series	ATMEGA328
Power supply		See Table 8 power supply	
Core		Cortex® M0+	megaAVR
Maximum frequency		48 MHz	20 MHz
flash memory		Up to 32 kBytes	32 kBytes
SRAM		Up to 12 kBytes	2 kBytes
EEPROM		Emulated in the flash memory(1)	1 kBytes
TIMER	General purpose	4	-
	Advanced	1	1
	Basic	-	1
ADC		1	1
DMA (number of independently configurable channels request)		3	-
USART		2	1
SPI		1	1
I^{2}C		1	1
I2S (Inter-IC-sound)		1	-
CRC		X	-
RTC		X	-
WWDG		X	X
IWDG		X	-
Comparotor		-(2)	X

1. *Refer to AN4894*
2. *STM32G0 series support this feature.*

5.2 System architecture

The STM32C0 series implement an Arm® 32-bit architecture with the M0+ core. While the ATMEGA328 family uses the Atmel AVR propriety core. The first one uses a Von Neumann architecture and the second one uses a Harvard architecture.

Table 7. **Comparison of CPU core**

Feature	M0+	megaAVR
Data path	32-bit	8-bit
Architecture	Von Neumann	Harvard
Pipeline	Two stages	One level
Instruction set	RISC	RISC
Program bus data width	32-bit	8-bit
Pre-fetch buffer	2 x 32-bit	16-bit
Debug interface	2-wire (SWD)	1-wire
ALU	-	YES
Cache instruction	16 bytes	-

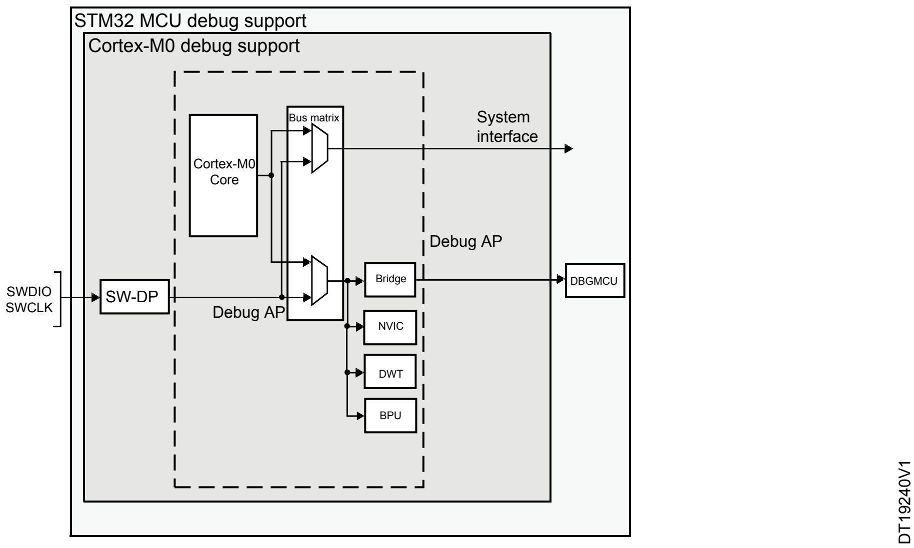
5.3 Debug

In addition to use the Arduino® API, it can be used the STM32 family Arm® debug tool. Two wires are needed to debug the MCU.

The new debug methodology allows:

- SW-DP: serial wire
- BPU: break point unit
- DWT: data watchpoint trigger
- Flexible debug pinout assignment
- NVIC debug
- MCU debug box (support for low-power modes, control over peripheral clocks, etc.)

Figure 3. **Block diagram of STM32C0 MCU and Arm® Cortex®-M0+-level debug support**



5.4 Power control peripheral

In the STM32C0 series, the PWR controller presents some differences versus ATMEGA328. The differences are summarized on the Table 8.

Table 8. Power control peripheral

PWR	STM32C0	ATMEGA328
Power supplies	• V_{DD}: 2.0 V to 3.6 V is the external power supply for the internal regulator and the system analog such as reset, power management and internal clocks • V_{DDA}: is the analog power supply for the A/D converter and shorted to V_{DD} due to the low number of pins • V_{DDIO}: is the power supply for the I/Os and shorted to V_{DD} due to the low number of pins	• V_{CC}: 2.7 V to 5.5 V is the digital supply voltage • AV_{CC}: 2.7 V to 5.5 V is the supply voltage pin for the A/D converter. It is mandatory to have it between V_{CC}-0.3 or V_{CC}+0.3
Power supply supervisor	• Integrated POR/PDR/BOR circuitry	• Integrated POR and BOR circuitry

5.5 Power consumption mode

The STM32C0 series and the STM32 family generally have different low-power mode. There are four different low modes: sleep, stop, standby and shutdown. The STM32C0 series has better dynamic and ultra-low power consumption than the ATMEGA328. Below there is a description of a different power consumption mode. All data is typical values contained in the datasheet (V_{DD}=3 V and temperature=25°C).

The dynamic consumption of the STM32C0 is better, for example the ATMEGA3238 has a power consumption at 8 MHz as much as the STM32C0 has a power consumption at 48 MHz.

The low-power modes use different names. But they have similarities, so it is easy to compare them.

Table 9. Power consumption comparison

Consumption mode	Clock	STM32C011/31	ATMEGA328	Unit
Run mode	8 MHz	0.655	4.2	mA
	4 MHz	0.365	1.2	
	1 MHz	0.225	0.2	
Idle/sleep mode	8 MHz	0.200	1.2	mA
	4 MHz	0.135	0.3	
	1 MHz	0.089	0.04	
Stop/power-save mode	Low speed clock on	74	4.2	µA
Standby/power down mode	All clock off	7450	900	nA
Shutdown mode	All clock off	19	-	nA

- Sleep mode corresponds to the ATMEGA328 idle mode. The CPU is clocked off, but other peripherals and interrupt controller continue to run. Moreover, in the STM32C0 is possible to have the memory.
- Stop mode is similar to the active power save mode. The high-speed clock is stopped, and the SRAM is retained. In both products it is possible to wake-up the product with a timer.
- Standby mode is similar to the power-down mode. The high-speed clock is off, the low-speed clock can be running if the application uses IWDG, but the main difference is that the RAM is powered off in the STM32C0.

- Shutdown mode has no equivalent in the ATMEGA328. It Is the ultimate low-power mode. All clocks and peripherals are off.
- Wake-up source: the Table 10 shows the wake-up source comparison.

Table 10. Wake-up source comparison

<table>
<tr><th>PWR</th><th>STM32C0</th><th>ATMEGA328</th></tr>
<tr><td>Low-power modes and wake-up sources</td><td>Sleep mode:
– Peripheral event/interrupt
– EXTI interrupt/event
– NVIC IRQ interrupt
– IWDG
– nReset
Stop mode:
– Peripheral event/interrupt
– EXTI interrupt/event
– NVIC IRQ interrupt
– IWDG
– nReset
Standby mode:
– Wake-up pins
– IWDG
– nReset
Shutdown mode:
– Wakeup pins
– nReset</td><td>Idle:
– Internal and external interrupts
– WDT (watch dog timer)
– Reset
ADC noise reduction:
– WDT
– Reset
– BOR reset
– Internal interrupt (TWI address match, Timer/Counter2 SPM/ EEPROM ready)
– External interrupt (INT0 or INT1)
Power-down:
– WDT
– Reset
– BOR reset
– Internal interrupt (TWI address match)
– External interrupt (INT0 or INT1)
Power-save:
– WDT
– Reset
– BOR reset
– Internal interrupt (TWI address match, Timer/Counter2)
– External interrupt (INT0 or INT1)
Standby:
– WDT
– Reset
– BOR reset
– Internal interrupt (TWI address match)
– External interrupt (INT0 or INT1)
Extended standby:
– WDT
– Reset
– BOR reset
– Internal interrupt (TWI address match, Timer/Counter2)
– External interrupt (INT0 or INT1)</td></tr>
</table>

5.6 Reset and clock controller (RCC) interface

5.6.1 Clocks

Table 11. **Clock configuration STM32C0 vs ATMEGA328**

RCC	STM32C0	ATMEGA328
HSI48	high-speed fully integrated RC oscillator at 48 MHz	-
HSI8	-	8 MHz internal RC oscillator
LSI	low-speed fully integrated RC oscillator at 32 kHz	128 kHz RC oscillator
HSE	4 to 48 MHz	0.4 to 16 MHz
LSE	32.768 kHz	32.768 kHz
System clock source	HSI48, LSI, HSE, LSE	HSI8, HSE, LSI, LSE
System clock frequency	• Up to 48 MHz • 12 MHz after reset based on HSI	• Up to 16 MHz • 1 MHz after reset
APB frequency	Up to 24 MHz	-
RTC clock source	LSI, LSE or HSE clocks divided by 32	-
Clock output	MCO (PIN1, PIN2): LSI, LSE, SYSCLK, HSI48, HSE LSCO (pin): LSI, LSE available in stop mode	CLKO (PIN): System clock
Internal oscillator measurement and calibration	Internal/external clock measurement inputs: • TIM14 inputs: GPIO, RTC, HSE/32, MCO, MCO2 • TIM16 inputs: GPIO, LSI, LSE, MCO2 • TIM17 inputs: GPIO, HSE/32, MCO, MCO2	-

Table 12. **High and low speed clock accuracy comparison**

Clock accuracy	Temperature	STM32C0	ATMEGA328
HSI factory calibrated	Full range	-2.5% to 2%	±14%
	0 to 85°C	±1%	-
	30°C	-0.37% to 0.83%	±2%
LSI	Full range	-7.8% to 3%	±40%

The STM32C0 has a good clock accuracy compared to the ATMEGA328. This can permit to clock other peripheral with the CCO output. HSI is used for USART communication.

5.6.2 Reset

In the STM32C0 series there are different types of resets.

The first one is the power reset, which set all register as their reset values excepted when exiting standby mode where the register outside VCORE domain are not impacted (back up registers, IWDG, standby/shutdown mode control).

The second one is the system reset, which reset all registers to their reset value excepted the reset flags, the RTC registers and the back-up registers.

The last one is the RTC domain reset, which only affects the RTC domains (LSE oscillator, RTC and RCC_CSR1 register).

Through specific option bits, the STM32C0 has also the possibility to change the configuration the NRST pin as:

- Reset input/output
- Reset input
- GPIO

Table 13. **Reset source comparison**

Reset source	STM32C0 series	ATMEGA328
Power-on Reset/Power-down reset	X(1)(2)	X
Brown-out reset	X(1)	X
Exit from standby mode	X(1)	-
Exit from shutdown mode	X(1)	-
Low level on the NRST pin	X(2)	X
WWDG reset	X(2)	-
IWDG reset	X(2)	X
Software reset	X(2)	-
Low-power mode security reset	X(2)	-
Option byte loader reset	X(2)	-

1. *Power reset*
2. *System reset*

5.7 Nested vectored interrupt controller (NVIC)

As both products have a different core. The interrupt handler is different. The STM32C0 devices use a nested vectored interrupt controller (NVIC). It is more advanced and therefore new feature like the trail chaining.

In the AVR core, when an interrupt occurs, all interrupts are disable by hardware whose means that it is not possible to have multiple interrupts.

Unlike the Cortex M0+ where it is possible to have multiple interrupts in the same. Thanks to tail chaining the program does not return to the main execution but jumps directly to the next interrupt, saving time by not recovering again the context saving of the main execution.

Figure 4. **Tail chaining**

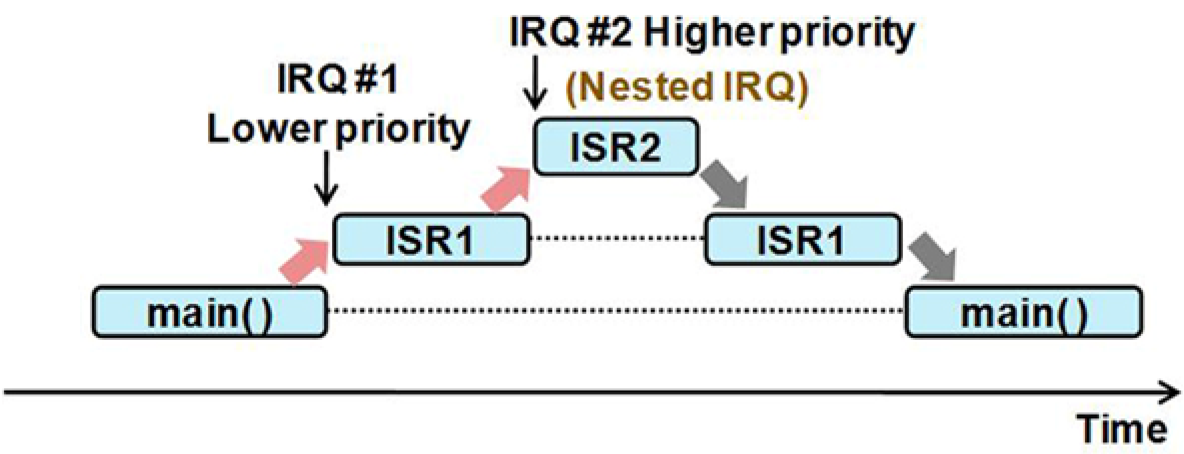


Table 14. **Interrupts features comparison**

Parameter	STM32C0	ATMEGA328
Interrupt vectors	Up to 32 interrupt vectors (+ 5 system ones)	Up to 26 interrupt vectors
Non-maskable interrupt	Yes	-
Interrupt priorities	4 levels lower number = higher priority	The lower the address the higher is the priority
Disable interrupts	yes, apart from NMI and HardFault	Yes
External interrupts	16 external interrupt channels linked to IO lines	2 external interrupts linked to two IOs 3 Pin change Interrupt linked to all IOs
Reset vector	4 bytes (address of the IRQ procedure)	3 bytes (boot reset address)
Interrupt latency	16 cycles to save context 16 cycles to restore context	4 cycles to save context 4 cycles to restore context
Tail chaining	Yes	No

The Cortex M0+ has six systems interrupt (five more than STM8). These interrupts are non-maskable, they are always activated. The priority of Reset, NMI and HardFault are fixed, in contrast to SVC, PendSV and SysTick, which are programmable.

Table 15. **System interrupts comparison**

Offset	STM32C0	ATMEGA328
0x00	-	**Reset** – address of the application start
0x04	**Reset** – address of the application start	-
0x08	**NMI**– Non Maskable Interrupt connected to SRAM parity error, HSE and LSE clock security systems (may be slightly different in other STM32 lines)	-
0x0C	**HardFault** – reports all isues related to bus/memory accesses	-
0x2C	**SVC** – system service call – software interrupt. Used by operating systems	-
0x38	**PendSV** – pendable request for system service software interrupt. Used by operating systems	-
0x3C	**SysTick** – interrupt from built in 24bit counter (part of the core) – used for delays, timeouts and operating system timing	-

5.8 DMA

Direct memory access (DMA) is a new IP in the STM32C0 compared to the ATMEGA328P. It is clearly a major asset to improve the product consumption and the CPU bandwidth. Data transfer without the CPU is allowed. It is between peripheral-to-memory, memory-to-peripheral, memory-to-memory and peripheral-to-peripheral.

In the STM32C0, the DMA has three different channels. Each channel has four levels of priority (very high, high, medium, and low), the channel one has priority over a request of channel two. Different transfer sizes are available (byte, half-word, and word). A circular buffer management can be used.

5.9 GPIO interface

The STM32C0 has seven configurations bits while the ATMEGA has only three bits. So, it can be possible:

- To choose between four different output speeds
- To have pull-up or pull-down
- To choose between seven GPIO configuration:
  – Input floating
  – Input pull-up
  – Input pull-down
  – Analog
  – Output open drain with pull-up or pull-down capability
  – Output push-pull with pull-up or pull-down capability
  – Alternate function push-pull with pull-up or pull-down capability
  – Alternate function open drain with pull-up or pull-down capability

They have also different V_{IH}/V_{IL} value due to the huge different between their power supply architecture.

Table 16. **GPIO input voltage comparison**

Voltage	STM32C0	ATMEGA328
V_{IL}	0 V to $0.3xV_{DD}$	-0.5 to $0.3xV_{DD}$
V_{IH}	$0.7xV_{DD}$ to 5 V	$0.6xV_{DD}$ to V_{DD}+0.5

5.10 RTC

The real time clock (RTC) is a new feature in STM32C0 that measure the time increasing. It can be clocked by LSE, LSI or the HSE (divided by a prescaler), but to have the best accuracy it needs to use the LSE clock source with 32.768 kHz external crystal. The RTC can measure the calendar date: sub-second, seconds, minutes, hours (12 or 24 format), weekday, date with an automatic correction for 28, 29 (leap year), 30, and 31 days of the month. It has one programmable alarm.

The RTC also can be used as timer to automatically wake up the product.

5.11 USART

Table 17. **USART peripheral**

Feature	STM32C0	ATMEGA328
Configurable oversampling method	16 or 8	No
Programmable data word length	7, 8 or 9 bits	5,6,7,8 or 9 bits
DMA	Continuous communications using DMA Received/transmitted bytes are buffered in reserved SRAM using centralized DMA.	-
Separate signal polarity control for transmission and reception	X	-
Swappable Tx/Rx pin configuration	X	-
Wake-up from low-power mode	X	-

5.12 I²C

Table 18. I²C configuration

Feature	STM32C0	ATMEGA328
Communication speeds	• Standard-mode (up to 100 kHz) • Fast-mode (up to 400 kHz) • Fast-mode Plus (up to 1 MHz)	Up to 400 kHz data transfer speed
Addressing mode	7-bit and 10-bit	7-bit
SMBus	3.0	-
PMBus	1.3	-
DMA capability	1-byte buffer	-
Clock choice	PCLK, SYSCLK, HSIKER	-

5.13 Flash memory

The STM32C0 has a maximum frequency at 48 MHz and the flash memory maximum frequency is 24 MHz. To compensate the flash memory speed and to be sure to have valid and uncorrupted data, a wait state feature has been added. It is not implemented on ATMEGA328 because the CPU speed does not go above the flash memory speed.

Table 19. Flash memory

Feature	STM32C0	ATMEGA328
Page size	• 2 kbytes	• 64 bytes
Security	• Read protection activated by option (RDP) • Two write protection areas, selected by option (WRP) • Two proprietary codes read protection areas, selected by option (PCROP) • Securable memory area	• Two different areas (read while write section and no read while write section) • Readout protection

5.14 SRAM

Table 20. SRAM density comparison

Maximum Flash memory density	STM32C0	ATMEGA328
16 Kbytes	STM32C011x4: 6 Kbytes STM32C031x4:12 Kbytes	-
32 Kbytes	STM32C011x6: 6 Kbytes STM32C031x6:12 Kbytes	2 Kbytes

5.15 Timer

Table 21. Timers available in STM32C0 series MCUs

Timer type	Timer	Counter resolution	Counter type	Prescaler factor	DMA request generation	Capture/ compare channels	Complementary outputs
Advanced control	TIM1	16-bit	Up, down, up/down	integer from 1 to 65536	YES	4 (6 internals)	3

Timer type	Timer	Counter resolution	Counter type	Prescaler factor	DMA request generation	Capture/ compare channels	Complementary outputs
General purpose	TIM3	16-bit	Up, down, up/down	integer from 1 to 65536	YES	4	-
	TIM14	16-bit	Up	integer from 1 to 65536	-	1	-
	TIM16 TIM17	16-bit	Up	integer from 1 to 65536	YES	1	1
Systick	STK	24-bit	Down	HCLK/8	-	-	-

All timers in the STM32C0 are 16-bit and the maximum clock is now 48 MHz and one 24-bit timer inside the Cortex M0+ core, which is generally used as 1 ms time base.

5.16 ADC

Table 22. ADC differences between STM32C0 series and ATMEGA328

Feature	STM32C0 series	ATMEGA 328
Resolution	12-bit, 10-bit, 8-bit, or 6-bit configurable	10 bits
Conversion time	Up to 0.4 µs (2.5Msps)	Up to 13 µs (76.9 ksps) and 15 ksps in full resolution
Self-calibration	YES	-
Programmable sampling time	YES	-
DMA support	YES	-
Conversion mode	Single + Continuous	Single
Oversampling	YES	-
Number of externa channel	Up to 19	8
Internal channel	• V_{sense} (temp sensor) • V_{refint} • V_{DDA} • V_{SSA}	• V_{sense} (temp sensor) • V_{refint}

5.17 SPI

Table 23. SPI comparison

Feature	STM32C0	ATMEGA328
Half-duplex synchronous transfer on two lines (with bidirectional data line)	X	-
Simplex synchronous transfers on two lines (with unidirectional data line)	X	-
Data size selection	4 to 16-bit data size selection	Only 8 bits
Multi-master mode capability	X	-
Faster communication - Maximum SPI speed	12 MHz	10 MHz
SPI Motorola support	X	-
DMA capability	Two 32-bit embedded Rx and Tx FIFOs	-
CRC feature	X	-

Feature	STM32C0	ATMEGA328
Enhanced TI and NSS pulse modes support	X	-
I^2S	X	-

Moreover, the STM32C0 is I^2S compatible. If the application needs an audio interphase.

5.18 Watchdog

The STM32C0 series has two different watchdogs: the independent watchdog (IWDG) and one the system window watchdog (WWDG). It is one more compared to the ATMEGA328.

Independent watchdog (IWDG)

The watchdog timer is integrated both in the ATMEGA328 and in the STM32C0. But in the ATMEGA328, it is not possible to define the counter reload value, the user can just choose between different oscillator dividers.

Nevertheless, in the STM32C0 series, this watchdog can be driven in two different ways:

- Without the window option activated the IWDG works. The counter value is reloaded when the key is written in IWDG_KR. And the chip reset, when the down counter value becomes lower than 0x000.
- With the window option activated the counter value can be reloaded by two different manners. With writing a special key in IWDG_KR. The second and new way is to refresh the counter value, by a value written in the window register. With this new feature it adds a new conditional reset in addition to the one described before. The circuit is reset if the down counter is reloaded outside the window.

Table 24. **IWDG comparison**

Feature	STM32C0 series	ATMEGA328
Clock source	LSI (32 kHz)	128 kHz
Down counter size	12-bit	NA
Window option	Yes	-
Minimum time out period	125 µs	16 ms
Maximum time out period	32.76 s	8.0 s
Debug mode (suspend the IWDG when the core is halted)	Yes	-
Freeze IWDG in low power mode	Yes (STOP and STANDBY)	-

System window watchdog (WWDG)

The STM32C0 has another watchdog dedicated to detecting the occurrence of a software fault. Usually this fault is generated by external interference or by unforeseen logical conditions, which causes the application program to give up its normal sequence. The watchdog circuits generates an MCU reset on expiry of a programmed time, unless the program refreshes the contents of the down-counter before the T6 bit becomes cleared.

Table 25. **WWDG**

Feature	STM32C0 series
Clock source	HSI, HSE, LSI, LSE
Static prescaler	4096
Variable prescaler	1-128
Minimum time out period for F_{WWDG}=16 MHz	0.512 ms
Maximum time out period for F_{WWDG}=16 MHz	2097.152 ms
WWDG interrupt	Yes
Debug mode (suspend the WWDG when the core is halted)	Yes

5.19 Option and engineering bytes

The fuse and lock bits of the ATMEGA328 can be equivalent of the option bytes present in the STM32C0. These bits are used to customize the configuration of the MCU like the flash protection, watchdog selection, lock the debug capability.

The STM32C0 series has a dedicated programming protocol with a locking mechanism to protect the option bytes from unwanted writes and they are complemented to maximize the robustness.

Table 26. Option bytes comparison

Feature	STM32C0 series	ATMEGA328
Register size	8x64-bits (32-bit option byte + 32-bit complemented option byte)	4x8-bits
Peripheral configuration	• NRST pin, reset holder, BOR and low power mode entry protection • Boot configuration • Multiple bonding • Clock remapping • Watchdog selection and freeze option • Flash protection (RDP, PCROP, WRP, SEC)	• BOR • Boot size and configuration • Watchdog selection • Memory protection

In addition to the option bytes, the STM32C0 has some engineering bytes that are visible to the user. They contain some useful information that is written during production testing, such as:

- Unique device ID
- Flash size
- Package type
- Calibration value of internal voltage reference and temperature sensor

6 Firmware migration using the Arduino® IDE

To migrate an Arduino® application from the ATMEGA328 family to the STM32C0 series, it is necessary to install STM32 Arduino® core and select the appropriate board as explained in Getting started. This core supports all Arduino® software APIs from ATMEGA328, and even more.

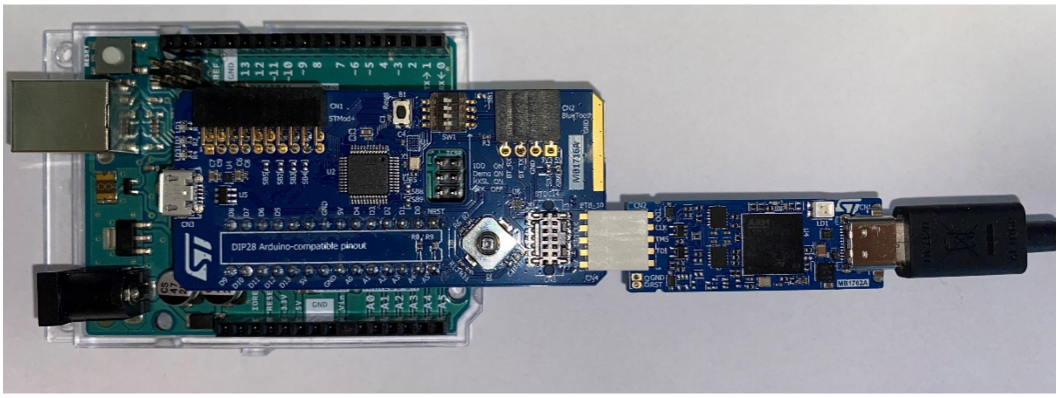
This means that all Arduino® build-in examples can be used, like blinky, AnalogReadSerial, toneMelody.

Furthermore, the external Arduino® libraries (for sensor, shields, ...) can be used if they respect Arduino® API's recommendation. If the application or the external libraries do not respect Arduino® API (for example with direct register access), then porting is necessary.

Firstly, the user must install STM32CubeProgrammer (available in STM32CubeProg).

To program the STM32C0 microcontrollers with the IDE and STM32CubeProgrammer the user needs to use the STLINK. Thus, with STM32C0316-DX it is necessary to use an external module like STLINK-V3 (MB1762) with the B2B connector.

Figure 5. Arduino® uno board with STM32C0316-DK and STLINK



To know more about the use of STM32C0 and Arduino® core see more details in STM32duino.

In addition to the Arduino® standard API, the STM32 Arduino® core provides other APIs to get access to some of STM32 hardware.

If needed, it is also possible to access to the STM32Cube drivers (HAL and LL) within Arduino® so that the application can access to full STM32 hardware capabilities.

Revision history

Table 27. Document revision history

Date	Version	Changes
19-Dec-2022	1	Initial release.

Contents

List of tables

List of figures